Puzzle Triangles for Littlewood–Richardson Coefficients

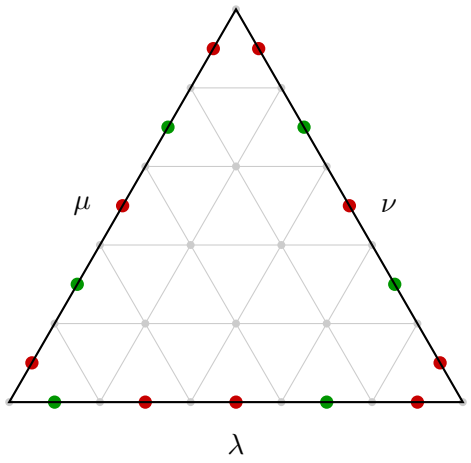
After Zinn-Justin, *LR coefficients and integrable tilings* (arXiv:0809.2392), Figure 8.

Encoding. A partition λ inside a $k \times m$ rectangle ($n = k + m$) is encoded as a binary string of length n: the i-th zero (green particle) is at position $m - \lambda_i + i$ for $i = 1, \ldots, k$. Remaining positions are ones (red particles). The LR coefficient $c^{\lambda}_{\mu,\nu}$ counts the number of valid tilings of the triangle with boundary conditions λ (bottom), μ (left), ν (right).

Example 1: $c^{(3,1)}_{(2,1),(2,1)} = 1$

All partitions in a 2×3 rectangle ($k = 2$, $m = 3$, $n = 5$).

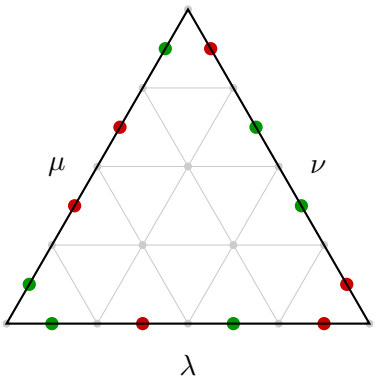
$\lambda = (3,1)$: zeros at $3-3+1 = 1$, $3-1+2 = 4$ $\quad \to$ `0 1 1 0 1`
$\mu = (2,1)$: zeros at $3-2+1 = 2$, $3-1+2 = 4$ $\quad \to$ `1 0 1 0 1`
$\nu = (2,1)$: zeros at $3-2+1 = 2$, $3-1+2 = 4$ $\quad \to$ `1 0 1 0 1`



Example 2: $c^{(2,1)}_{(2),(1,1)} = 1$

All partitions in a 2×2 rectangle ($k = 2$, $m = 2$, $n = 4$).

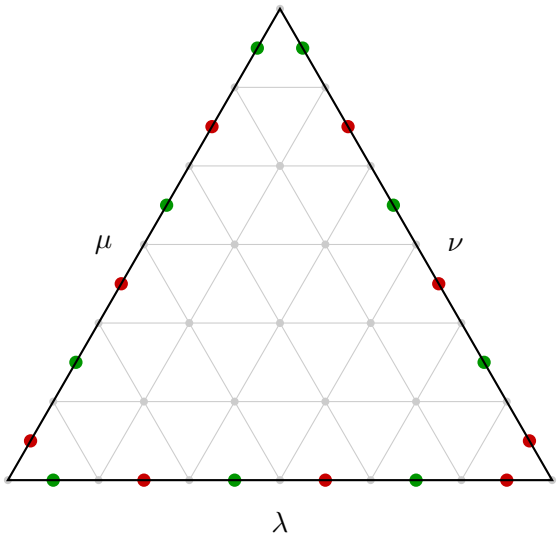
$\lambda = (2,1)$: zeros at $2-2+1 = 1$, $2-1+2 = 3$ $\quad \to$ `0 1 0 1`
$\mu = (2,0)$: zeros at $2-2+1 = 1$, $2-0+2 = 4$ $\quad \to$ `0 1 1 0`
$\nu = (1,1)$: zeros at $2-1+1 = 2$, $2-1+2 = 3$ $\quad \to$ `1 0 0 1`



Example 3: $c^{(3,2,1)}_{(2,1),(2,1)} = 2$

All partitions in a 3×3 rectangle ($k = 3$, $m = 3$, $n = 6$).

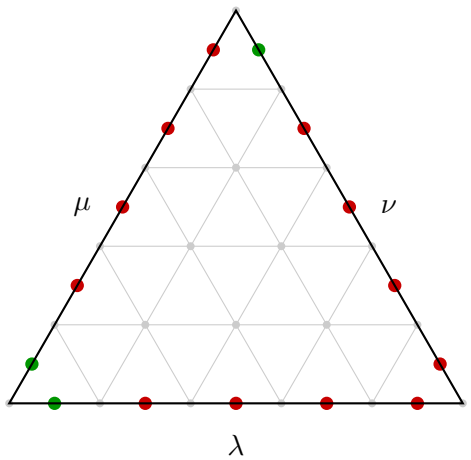
$\lambda = (3,2,1)$: zeros at $1, 3, 5$ $\rightarrow$ `0 1 0 1 0 1`
$\mu = (2,1,0)$: zeros at $2, 4, 6$ $\rightarrow$ `1 0 1 0 1 0`
$\nu = (2,1,0)$: zeros at $2, 4, 6$ $\rightarrow$ `1 0 1 0 1 0`



Example 4: Trivial case $c^{(4)}_{(4),\varnothing} = 1$

$\lambda = (4)$, $\mu = (4)$, $\nu = \varnothing$ in 1×4 rectangle ($k = 1$, $m = 4$, $n = 5$).

$\lambda = (4)$: zero at $4-4+1 = 1$ $\rightarrow$ `0 1 1 1 1`
$\mu = (4)$: zero at $4-4+1 = 1$ $\rightarrow$ `0 1 1 1 1`
$\nu = \varnothing = (0)$: zero at $4-0+1 = 5$ $\rightarrow$ `1 1 1 1 0`



The single green particle enters at the far left of the bottom, exits at the bottom of the left edge. All red particles travel from the bottom-right to the right edge.

Example 5: Pieri rule $c^{(3,1)}_{(2,1),(1)} = 1$

$\lambda = (3,1)$, $\mu = (2,1)$, $\nu = (1)$ in 2×3 rectangle ($k = 2$, $m = 3$, $n = 5$).

$\lambda = (3,1)$:	zeros at $1, 4$	$\to$ `0 1 1 0 1`
$\mu = (2,1)$:	zeros at $2, 4$	$\to$ `1 0 1 0 1`
$\nu = (1,0)$:	zeros at $3, 5$	$\to$ `1 1 0 1 0`

